

FRETGARDEN

Pedagogical Constitution

Chapter 7 - Review, Retention, and Spaced Retrieval

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This chapter defines how FretGarden protects learning after initial instruction. It governs when material returns, what learners must retrieve, how forgetting is interpreted, when re-teaching is required, and what evidence permits retention or reactivation claims.

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1. Purpose

This chapter defines the constitutional requirements for review, retention, spaced retrieval, reactivation, cumulative practice, and the interpretation of forgetting. It establishes how FretGarden must preserve core learning after the initial lesson and how later evidence may strengthen, weaken, or reactivate a capability claim.

Initial instruction creates an opportunity for learning. It does not guarantee that learning will remain available. Knowledge and skill change over time. Retrieval may become slower, representations may lose clarity, physical coordination may become less stable, and previously integrated capabilities may fragment when they are not used.

A coherent educational system must therefore design for forgetting rather than treat forgetting as an unexpected learner failure. Review is not an optional supplement added after the curriculum is complete. It is part of the curriculum architecture itself.

Every core capability MUST have a defined return path. A lesson that introduces important material without specifying how that material will be retrieved, reviewed, and revalidated is constitutionally incomplete.

This chapter does not prescribe one universal review calendar, one algorithm, or one fixed number of repetitions. It defines the educational obligations that any review system must satisfy.

2. Constitutional Authority

This chapter operationalizes the foundational commitments that core knowledge must be revisited after delay; retrieval is superior to passive re-exposure for demonstrating access; mastery claims must include retention appropriate to future use; progression must not rely indefinitely on stale evidence; and review must strengthen musicianship rather than application-specific familiarity.

Where curriculum pacing, engagement goals, or implementation convenience conflict with the need to preserve a required capability, the retention obligation governs. Review may be adapted in form, duration, representation, and scheduling, but it may not be silently omitted.

Later domain-specific chapters MAY define specialized review requirements for note knowledge, intervals, rhythm, technique, ear training, harmony, improvisation, and other domains. Those requirements must remain consistent with the principles established here.

3. Core Review Principle

Review exists to make prior learning available for future musical use, not merely to remind the learner that the material was once encountered.

A review activity is valid only when it serves a defined capability and produces interpretable evidence about that capability. Reopening a lesson, replaying an explanation, or displaying a diagram may support reorientation, but such exposure does not independently establish retention.

The central review sequence is:

- 1 attempt retrieval before complete re-exposure;
- 2 observe the quality and conditions of the attempt;
- 3 distinguish retrieval weakness from misconception, execution failure, or representational confusion;
- 4 provide only the support required to restore productive work;
- 5 require a corrected or reconstructed response;

- 6 vary or apply the capability when appropriate;
- 7 record what the evidence permits FretGarden to claim; and
- 8 schedule or trigger the next return according to educational need.

This sequence prevents review from becoming passive rereading and ensures that difficulty produces diagnostic information rather than only frustration.

4. Review, Re-Exposure, Re-Teaching, and Reactivation

4.1 Review

Review is a planned return to previously introduced material for retrieval, strengthening, integration, transfer, or revalidation. Review assumes that the learner has previously received meaningful instruction.

4.2 Re-Exposure

Re-exposure presents the material again without requiring sufficient independent access. It may be useful for orientation or confidence, but it provides weak retention evidence.

4.3 Re-Teaching

Re-teaching reconstructs the concept or skill because the learner cannot productively recover it through retrieval and limited support. Re-teaching is warranted when a prerequisite model has collapsed, a misconception is active, or prior instruction was insufficient.

4.4 Reactivation

Reactivation restores a capability that was previously demonstrated but has become unavailable, slow, fragmented, or context-bound. Reactivation is not the same as first instruction because prior learning may permit faster reconstruction.

4.5 Revalidation

Revalidation collects current evidence that a capability remains sufficiently available for an upcoming use, progression decision, or renewed mastery claim.

FretGarden MUST distinguish these functions. A learner who needs re-teaching should not be trapped in repeated retrieval failure, and a learner who only needs reactivation should not be forced through unnecessary full instruction.

5. Retention

5.1 Definition

Retention is the continued availability of a learned capability after a meaningful period in which the complete answer, model, or procedure has not been continuously present.

5.2 Retention Is Scoped

Retention claims must specify the capability, content scope, conditions, representation, support level, and delay. Retaining natural-note locations on two strings does not establish retention across the full fretboard. Retaining a scale shape does not necessarily establish retention of scale degrees or note content.

5.3 Retention Is Dimensional

A capability may be retained in one dimension and weakened in another. A learner may remember the correct notes but lose rhythmic stability, recall a formula but not locate it, or perform a shape while losing conceptual explanation.

5.4 Retention Is Not Permanence

No constitutional claim implies that knowledge can never weaken. Retention evidence supports a time-bounded inference. Critical capabilities may require later revalidation.

5.5 Retention Requires Retrieval Opportunity

A retention check must give the learner a fair opportunity to access the capability without complete reconstruction being supplied by the task itself.

6. Forgetting

Forgetting is a reduction in the availability, speed, precision, or integration of a previously learned capability. It is expected and must be interpreted educationally.

Forgetting may appear as:

- complete retrieval failure;
- slower access;
- increased dependence on cues;
- loss of detail while preserving the central relationship;
- confusion between related concepts;
- failure under variation;
- physical instability;
- rhythmic drift;
- or loss of integration while isolated components remain available.

FretGarden MUST NOT frame ordinary forgetting as moral failure, laziness, or proof that prior practice was worthless. The educational response is to identify what remains, restore what has weakened, and adjust future spacing or practice conditions.

Forgetting is evidence about the current state of learning and the adequacy of prior review. It is not evidence about the learner's worth or potential.

7. The Spacing Principle

Spacing distributes retrieval and practice across time. Its purpose is to require the learner to reconstruct access after some forgetting has occurred, thereby strengthening future availability and revealing whether the capability persists without immediate prompting.

Spacing is not defined by one universal interval. The appropriate return depends on:

- importance of the capability;
- initial learning strength;
- evidence consistency;

- prior forgetting rate;
- similarity to competing material;
- frequency of natural use;
- upcoming curriculum dependencies;
- physical versus conceptual demands;
- and the consequence of failure.

A review scheduled so early that the answer remains continuously active may measure short-term performance rather than durable access. A review scheduled so late that the learner cannot make a meaningful attempt may require re-teaching rather than productive retrieval.

Review timing SHOULD place the learner near the boundary of successful retrieval, not at the boundary of discouragement or uncontrolled guessing.

8. The Retrieval-First Rule

When a learner returns to previously taught core material, FretGarden SHOULD normally request an attempt before displaying the complete answer.

Retrieval-first review exists because an attempt reveals what remains available and strengthens access. Immediate re-exposure may create familiarity while concealing forgetting.

Retrieval-first does not require unsupported failure. The initial task must be appropriately scoped, clearly instructed, and accessible. When the learner cannot begin productively, support should be introduced in controlled steps.

Exceptions may include safety-critical technique, tasks where incorrect repetition could create harmful physical habits, accessibility needs, or cases where orientation to a changed representation is necessary before the capability can be demonstrated.

The amount of information shown before the attempt MUST be no greater than necessary to make the task valid and interpretable.

9. Retrieval Strength and Storage Strength

FretGarden distinguishes current accessibility from underlying learning history.

9.1 Retrieval Strength

Retrieval strength is the current ease with which a capability can be accessed under specified conditions.

9.2 Storage Strength

Storage strength is the inferred durability of the capability based on prior successful learning, varied use, delayed retrieval, and integration. It is not directly observed through one attempt.

A capability may have low current retrieval strength but substantial prior learning, allowing rapid reactivation. Another capability may feel easy because it was just practiced while remaining weakly stored.

FretGarden SHOULD avoid interpreting temporary ease as durable learning or temporary difficulty as complete loss. Review design should consider both the present attempt and the learner's longitudinal evidence.

10. Review Functions

Every review activity **MUST** have one or more declared functions.

10.1 Retrieval Strengthening

Increase access to previously learned knowledge or skill.

10.2 Retention Verification

Determine whether the capability remains available after delay.

10.3 Fluency Development

Reduce effort or response latency while preserving accuracy and adaptability.

10.4 Transfer Expansion

Apply the capability in a changed representation, key, position, tempo, or musical context.

10.5 Integration

Reconnect the capability to broader musical systems or dependent skills.

10.6 Error Repair

Correct a recurring misconception, execution fault, timing instability, or navigation weakness.

10.7 Reactivation

Restore a previously demonstrated capability after decay.

10.8 Readiness Revalidation

Confirm that a prerequisite remains available before consequential progression.

A review may serve several functions, but designers should identify the primary purpose so evidence and feedback remain interpretable.

11. Review Unit Architecture

A review unit is a bounded return to prior learning. It may be embedded inside a new lesson, appear as a standalone session, or occur within cumulative practice.

A compliant review unit **SHOULD** contain the following functions:

- 1 identify the capability and reason for return;
- 2 establish the current conditions and permitted support;
- 3 request an initial retrieval or performance attempt;
- 4 interpret accuracy, latency, confidence, prompting, and error type;
- 5 select continuation, support, reactivation, or re-teaching;
- 6 require corrected retrieval or performance;
- 7 introduce variation or application when appropriate;
- 8 record evidence and claim limits; and

9 define the next return condition.

The visible form may be brief. A thirty-second retrieval check may satisfy the architecture when the capability is narrow and stable. A weakened integrated skill may require a longer reactivation sequence.

12. Initial Review Timing

The first return after instruction is particularly important because it reveals whether the learner can reconstruct the capability after attention has shifted.

Core material SHOULD normally be retrieved:

- within the original lesson after intervening activity;
- again after the lesson or session has ended;
- and again after a longer delay appropriate to the capability.

These are conceptual stages rather than universal clock intervals. Domain-specific chapters may define typical windows.

An initial return that requires complete re-teaching suggests that the original instruction, practice amount, scaffold fading, or exit evidence may have been insufficient. The review system should not treat all such failures as scheduling problems.

13. Adaptive Review Timing

Review timing SHOULD respond to evidence rather than remain identical for every learner and capability.

A return may be scheduled sooner when:

- initial performance was heavily prompted;
- accuracy was inconsistent;
- the capability is a near-term prerequisite;
- the learner confused related concepts;
- prior retention checks failed;
- or physical execution remained unstable.

A return may be scheduled later when:

- independent retrieval was stable;
- the capability transferred across contexts;
- the learner used it naturally in later work;
- several delayed checks were successful;
- or the capability is strongly integrated with frequently used knowledge.

Longer spacing must not be awarded merely because a learner answered quickly once. Evidence strength, not convenience or reward logic, governs spacing decisions.

14. Expanding Review Intervals

When repeated evidence shows stable retention, review intervals SHOULD generally expand. Expanding intervals reduce unnecessary repetition while continuing to test durability.

Expansion should be based on:

- independent success;
- multiple observations;
- delayed retrieval;
- variation;
- low error recurrence;
- and successful use in dependent musical tasks.

Intervals should contract when evidence weakens. Contraction is an educational adjustment, not a demotion or punishment.

A capability that is naturally exercised often may require fewer dedicated review tasks, provided natural use genuinely requires retrieval of the target capability rather than bypassing it through habit or external support.

15. Cumulative Review

Cumulative review combines material from multiple prior lessons or units. Its purpose is to preserve important capabilities, reveal interference, and support integration.

Cumulative review **MUST** be intentionally sampled. It must not become an indiscriminate collection of old questions.

Selection should prioritize:

- foundational prerequisites;
- capabilities with evidence of decay;
- relationships that are easily confused;
- skills required by upcoming material;
- high-transfer concepts;
- and representative material from the learner's broader knowledge network.

Cumulative review should remain proportionate. Excessive review can crowd out new learning, while insufficient review allows prerequisite decay to accumulate invisibly.

16. Interleaving

Interleaving mixes related concepts or skill types so the learner must identify which knowledge or strategy applies. It differs from blocked practice, in which one type is repeated continuously.

Interleaving is useful for discrimination, strategy selection, and transfer. It is not universally appropriate during first contact.

Interleaving **SHOULD** begin only after each included capability has sufficient initial clarity to make selection meaningful.

Interleaved review must be designed so that difficulty comes from choosing among relevant alternatives, not from unrelated clutter or excessive task switching.

Examples include:

- mixing natural-note locations across strings;

- alternating interval types;
- identifying chord tones within changing harmonies;
- switching rhythmic subdivisions;
- or selecting an appropriate practice strategy for different error patterns.

17. Blocked Review

Blocked review repeats one capability or closely related item set. It is appropriate when restoring a weakened model, stabilizing execution, isolating a misconception, or building initial confidence.

Blocked review becomes insufficient when the learner succeeds through temporary repetition patterns rather than genuine selection or retrieval.

A common sequence is:

- 1 brief blocked reactivation;
- 2 reduced support;
- 3 interleaved discrimination;
- 4 variation;
- 5 and musical application.

The transition should occur when the learner demonstrates enough clarity that additional identical repetition provides diminishing educational value.

18. Desirable Difficulty in Review

Review should require meaningful effort without making success primarily dependent on guessing or reconstruction from scratch.

Appropriate review difficulty may be created by:

- increasing delay;
- reducing cues;
- varying roots, positions, or representations;
- mixing related concepts;
- requiring generation rather than recognition;
- adding musical context;
- or increasing temporal demands after accuracy is stable.

Difficulty is excessive when the learner cannot identify a productive strategy, when errors cease to be interpretable, or when several unstable prerequisites fail simultaneously.

Effort is valuable only when it remains connected to the target capability and permits correction.

19. Retrieval Cues

A retrieval cue is information that narrows or activates access without supplying the complete answer.

Possible cues include:

- a root note;
- a first step in a procedure;
- a partial diagram;
- an interval category;
- a count-in;
- a tonal reference;
- or a reminder of the relevant strategy.

Cue use **MUST** be recorded when it materially changes the claim. Success after a cue may support guided or reconstructed capability but not necessarily independent retention.

Cues should be introduced progressively when the learner cannot retrieve independently. The system should use the least revealing cue likely to restore productive effort.

20. Recognition in Review

Recognition tasks may support early reorientation, discrimination, and efficient sampling. They are weaker than constructed response for claims about recall.

Recognition review is appropriate when:

- the intended future use is recognition;
- the task tests subtle perceptual distinctions;
- a broad set must be sampled efficiently;
- or it precedes a stronger retrieval task.

Recognition should not dominate review of knowledge expected to be generated independently. Selected-response success must account for chance and process-of-elimination strategies.

A review system **MUST NOT convert recognition accuracy into a free-recall or mastery claim without additional evidence.**

21. Generation and Reconstruction

Generation requires the learner to produce an answer, sound, location, pattern, explanation, or strategy. Reconstruction requires rebuilding a relationship from component knowledge.

These forms of retrieval provide strong evidence because they expose the learner's internal model.

Examples include:

- naming a note without answer choices;
- constructing an interval from a root;
- rebuilding a chord formula;
- playing a rhythm from notation;
- drawing a partial fretboard pattern;
- or explaining how to repair an unstable passage.

Generation tasks must remain scoped. Excessive writing, typing, or interface demands must not contaminate the musical capability being reviewed.

22. Context Variation

A capability retained only in the exact practiced context is fragile.

Review SHOULD vary incidental features while preserving the underlying relationship. Variation may include:

- root;
- key;
- string set;
- fretboard region;
- register;
- tempo;
- rhythmic order;
- notation system;
- instrumental context;
- or musical phrase.

Variation must be meaningful but controlled. Changing several unfamiliar variables at once can obscure the source of failure.

A review sequence should distinguish near transfer from broader integration so evidence claims remain appropriately scoped.

23. Musical Application in Review

Review should eventually return capabilities to musical use. Isolated retrieval confirms access; application confirms that access can support musicianship.

Application may include:

- using chord tones in a progression;
- recognizing a scale degree in a melody;
- applying a rhythmic subdivision to a riff;
- constructing a voicing in a new key;
- or choosing a practice method for a real passage.

Application tasks should not be the only review evidence when several components could mask one another. Controlled retrieval and musical application serve complementary purposes.

24. Review of Physical Skills

Physical skills require review because coordination, timing, tone, and tension can decay even when conceptual knowledge remains.

Physical review must begin at conditions that permit control. The learner should not be required to reproduce a historical peak tempo before current stability is verified.

Review should assess relevant dimensions such as:

- movement accuracy;

- tone;
- noise control;
- timing;
- tension;
- repeatability;
- transition quality;
- and recovery after disruption.

When a physical pattern has weakened, reactivation may require reduced tempo, smaller chunks, or renewed modeling. Speed restoration follows control rather than precedes it.

25. Review of Temporal Skills

Rhythmic and metronome-related capabilities must be reviewed as relationships to pulse, not merely as remembered note sequences.

Review may include:

- counting and clapping;
- maintaining pulse through rests;
- subdivision with reduced click frequency;
- performing familiar material at changed tempos;
- identifying timing errors;
- or transferring a rhythmic structure to new pitches.

A learner who remembers the notes but loses temporal organization has not retained the complete musical capability.

26. Review of Perceptual and Ear-Training Skills

Perceptual skills require varied examples because repeated exact recordings may be memorized as items rather than recognized as relationships.

Review should vary roots, registers, timbres, voicings, and contexts within the learner's readiness.

Perception evidence is strengthened when the learner can:

- classify the sound;
- reproduce it;
- predict it;
- locate it;
- or explain its function.

Visual information that reveals the answer must not be present when the review claim concerns hearing.

27. Review of Conceptual Knowledge

Conceptual review must require more than repeating definitions.

Appropriate review may ask the learner to:

- derive an answer;
- compare related structures;
- correct a misconception;
- predict the result of a change;
- apply a formula to a new root;
- or explain a musical consequence.

Definitions may be reviewed, but the larger objective is an organized model that remains useful under variation.

28. Review of Fretboard Navigation

Fretboard review must prevent dependence on one memorized position, root, or highlighted diagram.

A balanced review sample may include:

- absolute note locations;
- relative intervals;
- scale degrees;
- chord tones;
- connections between positions;
- multiple anchors;
- and navigation within musical context.

The system should vary starting points and reduce visual support. Shape execution alone must not be used to infer note, interval, or functional retention.

29. Review of Metacognitive Practice Skills

Practice judgment also requires retention. Learners should revisit how to set targets, regulate difficulty, respond to errors, and decide when to stop or progress.

Review evidence should involve actual decisions. For example, the learner may be asked to choose a tempo, diagnose a failure, or design a short practice sequence.

Repeatedly displaying practice advice does not establish that the learner can use it independently.

30. Review Evidence

Review evidence must record enough context to support interpretation.

Relevant dimensions include:

- time since last instruction or successful retrieval;
- prompting level;
- accuracy;
- response latency or tempo where relevant;
- number and pattern of errors;
- self-correction;

- context familiarity;
- representation;
- transfer demand;
- and whether re-teaching occurred before success.

A correct answer after the complete solution was shown is evidence of corrected immediate performance, not independent retention.

Evidence records SHOULD preserve distinctions rather than compress all review outcomes into one percentage.

31. Retention Claims

A retention claim is permitted only when the learner demonstrates the capability after a meaningful delay with support appropriate to the claim.

Retention claims may be qualified as:

- retained with cues;
- independently retained in familiar conditions;
- retained across variation;
- retained and fluent;
- or retained and integrated in musical use.

The delay must be interpreted relative to the capability and prior evidence. Later standards may define domain-specific windows.

A review task MUST NOT claim retention when it immediately re-teaches the answer before measuring the learner's access.

32. Review Outcomes

A review may produce one of the following educational outcomes:

32.1 Confirmed

The capability remains available at the required level.

32.2 Strengthened

The capability was available and became more stable, fluent, or transferable through review.

32.3 Partially Retained

Some dimensions remain available while others require work.

32.4 Reactivated

The capability was initially weak or unavailable but returned through limited support and corrected retrieval.

32.5 Re-Teaching Required

The learner could not restore the capability through reasonable cues or brief reconstruction.

32.6 Misconception Detected

Review revealed a coherent incorrect model requiring targeted correction.

32.7 Evidence Inconclusive

Technical, accessibility, instruction, or sampling conditions prevented a valid inference.

These outcomes should guide the next educational action rather than serve as moral labels.

33. Reactivation Architecture

Reactivation should restore prior learning with the least unnecessary repetition.

A typical reactivation sequence is:

- 1 identify what remains available;
- 2 provide a minimal cue or anchor;
- 3 reconstruct the missing relationship;
- 4 perform or retrieve correctly;
- 5 repeat under reduced support;
- 6 vary the context;
- 7 and schedule a nearer return.

Rapid reactivation provides evidence that prior learning remains partly stored, but it does not by itself restore a prior mastery claim. Delayed independent evidence may still be required.

34. Re-Teaching Threshold

Review should transition to re-teaching when continued retrieval attempts no longer produce useful evidence or improvement.

Indicators include:

- complete inability to begin;
- repeated uncontrolled guessing;
- a stable misconception;
- failure across multiple cue levels;
- physical execution that cannot be corrected safely;
- or prerequisite collapse.

Re-teaching should target the missing model or skill rather than repeat the entire original lesson by default.

After re-teaching, the capability returns to an acquisition and stabilization cycle before later retention can be claimed.

35. Review and Progression

Review evidence may affect readiness and progression when the reviewed capability is a required prerequisite.

A lapse does not automatically require stopping all new learning. The response depends on:

- importance of the prerequisite;
- severity of the lapse;
- whether the new lesson can provide natural reactivation;
- risk of compounding misconception;
- and availability of support.

Possible responses include:

- continue with embedded review;
- advance with targeted support;
- pause for reactivation;
- route to remediation;
- or require revalidation before progression.

Progression rules must remain evidence-based and proportionate.

36. Review Load

Review load is the amount of prior material scheduled within a period of learning.

Review load must be sufficient to preserve core learning but not so large that learners spend most available time maintaining low-value details.

Prioritization should consider:

- foundational importance;
- frequency of future use;
- transfer value;
- evidence of decay;
- learner goals;
- and cost of reconstruction if forgotten.

Supporting or enrichment content may receive lighter review obligations than core content. Not every fact deserves permanent high-frequency retrieval.

37. Review Fatigue and Diminishing Attention

Review can become ineffective when attention declines, responses become automatic without thought, or the learner experiences repetitive overload.

Signs include:

- careless errors increasing over time;
- rapid guessing;
- declining rhythmic or physical control;
- irritability or disengagement;
- and correct responses based on item sequence rather than capability.

The appropriate response may be to stop, vary the task, take a break, reduce the set, or move to musical application.

More review is not automatically better. Review must preserve attention and diagnostic value.

38. Learner Agency in Review

Learners should gradually participate in review planning.

They may be taught to identify:

- which capabilities feel unavailable;
- which errors recur;
- which knowledge is needed for upcoming music;
- whether a task requires review or re-teaching;
- and when a practice strategy should change.

Self-selected review should complement, not completely replace, evidence-based scheduling. Learners may avoid weak areas or overpractice comfortable material without guidance.

FretGarden should make the reason for review understandable without requiring the learner to manage the entire system alone.

39. Learner-Facing Language

Review language must remain accurate, calm, and nonpunitive.

Appropriate language includes:

- “This skill is ready for another retrieval check.”
- “You remembered the structure but need to rebuild the note locations.”
- “This was reactivated with a cue; we will return to it sooner.”
- “Your timing remained stable after the delay.”
- “This prerequisite needs a short refresh before the next lesson.”

Inappropriate language includes claims that the learner failed because they forgot, lost all progress, broke a streak, or must repeat work as punishment.

Review should be framed as maintenance, strengthening, and re-access rather than proof that earlier learning was wasted.

40. Review Metrics and Their Limits

Metrics may summarize review behavior, but they must not replace educational interpretation.

Potentially useful metrics include:

- independent retrieval accuracy;
- cue dependence;
- response latency;
- retention interval;

- error recurrence;
- reactivation speed;
- and transfer success.

Potentially misleading metrics include:

- total review questions completed;
- time spent without quality information;
- streak length;
- raw repetitions;
- or a single blended retention score that conceals domain differences.

Metrics must not encourage unnecessary reviewing, avoidance of difficult material, or rapid low-quality responses.

41. Review Sampling

A review system cannot revisit every item equally. Sampling must represent the objective scope and prioritize educational risk.

Sampling SHOULD account for:

- coverage across strings, roots, keys, and positions;
- easy and difficult items;
- common confusions;
- recent errors;
- previously weak contexts;
- and examples not seen during immediate practice.

Sampling must not repeatedly select a narrow subset and then generalize performance to the whole capability.

42. Interference

Interference occurs when related learning competes during retrieval. Examples include confusing major and minor thirds, root and tonic, adjacent note names, or similar rhythmic patterns.

Review should use controlled contrast to strengthen discrimination.

When interference appears, FretGarden SHOULD:

- 1 identify the competing concepts;
- 2 clarify the defining distinction;
- 3 practice each accurately in controlled conditions;
- 4 contrast them directly;
- 5 interleave them;
- 6 and verify transfer after delay.

Repeated exposure without explicit contrast may strengthen confusion rather than resolve it.

43. Review After Mastery

Mastery does not end review obligations when the capability remains important.

After mastery, review may become:

- less frequent;
- more integrated;
- more varied;
- embedded in authentic music;
- or triggered by upcoming use.

A mastered capability should not remain permanently isolated in beginner-style drills if later musical use can maintain it more meaningfully.

However, natural use must genuinely exercise the capability. Repeatedly playing one memorized song may not maintain broad note recall or transfer.

44. Review After Failure

A failed review attempt should produce an educational decision, not only a lower score.

The next step depends on the error:

- a cue for retrieval weakness;
- contrast for confusion;
- reduced tempo for execution instability;
- counting for temporal failure;
- reconstruction for conceptual gaps;
- or re-teaching for collapsed prerequisites.

Repeatedly presenting equivalent tasks without modifying support or instruction is not adequate remediation.

45. Review Across Representations

Core musical knowledge should gradually remain available across relevant representations.

A learner may review the same underlying relationship through:

- sound;
- fretboard location;
- note names;
- interval numbers;
- scale degrees;
- tablature;
- standard notation;
- or musical performance.

Cross-representation review strengthens integration, but representation changes must not be introduced as hidden confounds. The learner should first understand how the representations map to one another.

46. Accessibility and Equivalent Review Evidence

Review must permit equivalent demonstrations when a standard response mode is inaccessible and the alternate method preserves the intended capability.

For example, a learner unable to provide a particular motor response may demonstrate conceptual or auditory retention through another valid method. However, an alternate response cannot substitute for a physical-execution objective when physical execution itself is the construct being assessed.

Accessibility adaptations should modify irrelevant barriers while preserving the educational claim.

47. Review Anti-Patterns

The following practices are constitutionally incompatible or require strong justification:

- showing the complete answer before every retention attempt;
- equating lesson reopening with review;
- using one fixed schedule for all capabilities regardless of evidence;
- punishing forgetting with excessive repetition;
- counting recognition as recall;
- reviewing identical items until sequence familiarity replaces retrieval;
- increasing speed before accuracy returns;
- allowing review queues to grow without prioritization;
- treating every lapse as full loss of mastery;
- or using streak preservation as the primary reason to review.

48. Constitutional Non-Negotiables

- 1 Every core capability **MUST** have a planned or condition-triggered return path.
- 2 Review **MUST** serve a defined objective and produce interpretable evidence.
- 3 Retention **MUST NOT** be inferred from passive re-exposure.
- 4 Core review **SHOULD** normally request retrieval before complete answer presentation.
- 5 Recognition **MUST NOT** be interpreted as free recall without additional evidence.
- 6 A delayed check **MUST** occur before a retention claim is made.
- 7 Review timing **MUST** consider capability importance and evidence strength.
- 8 Forgetting **MUST** be treated as educational information rather than learner misconduct.
- 9 Review failure **MUST** produce guidance, support, reactivation, or re-teaching.
- 10 Identical blocked repetition **MUST** eventually give way to variation, interleaving, or application when the objective requires transfer.
- 11 Review evidence **MUST** record prompting and re-teaching that materially affect interpretation.
- 12 Reactivation **MUST NOT** automatically restore a prior mastery claim without sufficient current evidence.

- 13 Review load MUST be prioritized and sustainable.
- 14 Physical and temporal skills MUST be reviewed under conditions that preserve control and safety.
- 15 Learner-facing review language MUST remain accurate and nonpunitive.
- 16 Implementation convenience MUST NOT justify omitting review of required prerequisites.

49. Conformance Tests

A review system or activity conforms to this chapter only if reviewers can answer:

- 1 What capability is returning?
- 2 Why is it returning now?
- 3 What evidence existed before this return?
- 4 What is the primary review function?
- 5 What will the learner attempt before seeing the complete answer?
- 6 What support is permitted?
- 7 How will guessing or item familiarity be controlled?
- 8 What delay is being tested?
- 9 Which quality dimensions matter?
- 10 How will the result distinguish retained, partially retained, reactivated, and re-teaching-required states?
- 11 What variation or application is appropriate?
- 12 How will the outcome affect future review timing?
- 13 How will it affect progression or prerequisite confidence?
- 14 What learner-facing claim is permitted?
- 15 And how does this review contribute to durable musical independence?

50. Tradeoffs and Resolution Rules

50.1 Review Versus New Learning

Review preserves foundations, while new learning expands capability. Neither should permanently displace the other. Required prerequisites and evidence of decay receive priority, but low-value detail should not consume disproportionate time.

50.2 Productive Difficulty Versus Confidence

Review should require effort, but repeated unproductive failure damages attention and provides little benefit. Minimal cues and reactivation preserve challenge while restoring a path to success.

50.3 Standardization Versus Adaptation

Shared review principles support coherence, but timing and task form must adapt to domains and learner evidence. Standards govern decisions rather than impose one universal schedule.

50.4 Efficient Sampling Versus Comprehensive Evidence

Brief review is practical, but broad claims require representative evidence. Sampling intensity should match the importance and consequence of the claim.

50.5 Natural Use Versus Dedicated Review

Authentic musical use can maintain capabilities, but only when the target relationship is actually retrieved. Dedicated review remains necessary when natural use is narrow, heavily prompted, or insufficiently diagnostic.

50.6 Learner Choice Versus Required Maintenance

Learners should gain agency, but required prerequisites cannot depend entirely on preference. The system may permit choice within a constitutionally necessary review obligation.

51. Assumptions Made

- 1 FretGarden will distinguish lesson completion from current capability status.
- 2 Core objectives will be identifiable at a granularity that permits review.
- 3 Review may occur inside lessons, between lessons, or within dedicated sessions.
- 4 Some evidence will be collected longitudinally.
- 5 Different domains will require different review forms and intervals.
- 6 Learners will sometimes forget previously demonstrated material.
- 7 Natural musical use may contribute valid review evidence when the target capability is observable.
- 8 Later chapters will define domain-specific timing recommendations and scoring details.
- 9 Accessibility may require alternate response methods.
- 10 And review systems will remain subordinate to the constitutional goals of durable, transferable musicianship.

52. Unresolved Questions

- 1 What default review windows should apply to major objective categories?
- 2 How should evidence from natural musical use modify dedicated review scheduling?
- 3 What minimum evidence should permit expansion to very long intervals?
- 4 How should learner confidence affect, but not control, review selection?
- 5 When should a lapsed mastery designation be formally downgraded rather than marked for revalidation?
- 6 How should physical-technique review account for changes in instrument setup or physical condition?
- 7 What review limits should prevent queue overload?
- 8 How should open-ended creative capabilities be reviewed over time?
- 9 Which forms of automated scheduling and evaluation are sufficiently trustworthy?
- 10 And how should review obligations differ between core pathways and optional enrichment content?

53. Topics Intentionally Deferred

- feedback taxonomy and wording;
- formal error classification;
- remediation architecture;
- domain-specific review intervals;
- adaptive sequencing algorithms;
- educational analytics;
- practice logging;
- ear-training scheduling details;
- rhythm and metronome review protocols;
- technique revalidation;
- creative-work review rubrics;
- accessibility standards;
- and implementation-specific review queues or notifications.

54. Required Future Cross-References

54.1 Feedback and Error Classification

Must reference review outcomes, retrieval failure, interference, partial retention, cue use, and re-teaching thresholds.

54.2 Remediation and Adaptive Difficulty

Must reference reactivation architecture, re-teaching thresholds, review failure, and contracted spacing.

54.3 Ear-Training Pedagogy

Must define varied auditory review, guessing controls, and perceptual retention evidence.

54.4 Rhythm and Metronome Pedagogy

Must define review of pulse, subdivision, synchronization, silent beats, and tempo restoration.

54.5 Fretboard Pedagogy

Must define representative sampling across strings, roots, positions, functions, and representations.

54.6 Practice-Session Design

Must define sustainable review load, cumulative review placement, and learner-selected maintenance.

54.7 Educational Analytics

Must preserve distinctions among retrieval, cue dependence, reactivation, re-teaching, retention interval, and transfer.

54.8 Constitutional Governance

Must define how changes to review standards affect existing mastery and progression records.

55. Chapter Conformance Statement

A FretGarden review system conforms to this chapter only when it creates a defensible path from prior learning to current retrieval evidence, interprets forgetting without punishment, provides proportionate support, distinguishes reactivation from re-teaching, and schedules future returns according to educational need.

A capability is not retained merely because it was once mastered, recently viewed, or recognized among answer choices. Retention must be demonstrated after delay under conditions appropriate to the intended future use.

Review is constitutionally complete only when it strengthens access, clarifies the current state of learning, produces an appropriate next action, and preserves the learner's movement toward independent musicianship.

FretGarden does not review material to keep learners busy. It reviews material so that what has been learned remains available when music requires it.

End of Chapter 7 - Review, Retention, and Spaced Retrieval